

Fruit Fly

Team: Cheesecake

Element K: Reflection on the Design

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

We identified our problem and design requirements, brainstormed, constructed a prototype, tested our prototype, redesigned a new prototype. Our problem and design requirements were identified by a stakeholder that asked us to create a solution. We also acquired requirements from other involved stakeholders to base a solution design on. Next, we brainstormed multiple ideas based on the requirements given to us. We selected one solution that fit the requirements the best and also was a better solution than any current solution that was in practice. After we selected a solution, we constructed a prototype using a 3D printer. During our testing phase where we saw if the prototype performed at the expected level of the requirements, we identified aspects that could be improved or altered. These parts of the first prototype were used to design and construct a second prototype that aligns with the design requirements more closely.

We selected our idea not only based on our physical design requirements but also its cost in the form of monetary expenditure, labor, and time. The testing process was very straightforward and gave us the information needed to be confident in our prototype.

During the brainstorming process, it is very easy to try to judge your ideas when they are thought of and then dismissed before they are illustrated or described on a page. Instead of dismissing ideas, we learned to write all of our ideas down not only because they could be thought of differently from person to person, but also ideas can be changed or combined in some form before they are materialized.